pressure will drive any star in the shaded region to explode. In the region to the right of the white-dwarf curve (white region; stars with larger circumferences), gravity completely overwhelms the star's degeneracy pressure. Any *cold* star which finds itself in this region will immediately implode under gravity's squeeze.

The Sun can live in the white region only because it is now very hot; its thermal (heat-induced) pressure manages to counterbalance its gravity. However, when the Sun ultimately cools down, its thermal pressure will disappear and it no longer will be able to support itself. Gravity will force it to shrink smaller and smaller, squeezing the Sun's electrons into smaller and smaller cells, until at last they protest with enough degeneracy pressure (nonthermal pressure) to halt the shrinkage. During this shrinkage "death," the Sun's mass will remain nearly constant, but its circumference will decrease, so it will move leftward on a horizontal line in Figure 4.4, finally stopping on the white-dwarf curve—its grave. There, as a white dwarf, the Sun will continue to reside forever, gradually cooling and becoming a black dwarf—a cold, dark, solid object about the size of the Earth but a million times heavier and denser.

This ultimate fate of the Sun seemed quite satisfactory to Eddington. Not so.the ultimate fate of a star more massive than Chandrasekhar's white-dwarf limit of 1.4 solar masses-for example, Sirius, the 2.3-solar-mass companion of Sirius B. If Chandrasekhar were right, such a star could never die the gentle death that awaits the Sun. When the radiation it emits into space has carried away enough heat for the star to begin to cool, its thermal pressure will decline, and gravity's squeeze will make it shrink smaller and smaller. For so massive a star as Sirius, the shrinkage cannot be halted by nonthermal degeneracy pressure. This is clear from Figure 4.4, where the shaded region does not extend high enough to intercept Sirius's shrinking track. Eddington found this prediction disturbing.

"The star has to go on radiating and radiating and contracting and contracting," Eddington told his audience, "until, I suppose, it gets down to a few kilometers radius, when gravity becomes strong enough to hold in the radiation, and the star can at last find peace." (In the words of the 1990s, it must form a black hole.) "Dr. Chandrasekhar had got this result before, but he has rubbed it in in his last paper; and, when discussing it with him, I felt driven to the conclusion that this was almost a *reductio ad absurdum* of the relativistic degeneracy formula. Various accidents may intervene to save the